IMDb TOP 700:
DECODING WHAT MAKES A MOVIE TIMELESS

Exploratory Data Analysis and Recommendations for the IMDb Content Insights Team

Graduation Project | September 2026

1. Project Background

IMDb ranks films using a Bayesian-weighted average rather than a raw average score. The weighting pulls films with few votes towards the overall mean, so a film can only rank high if it has both a strong rating and enough votes to back it. Studios, streaming platforms and critics use this ranking as a signal of a film's lasting reputation.

IMDb has extended this method to a Top 700 list. The Content Insights team wants to go beyond listing ranked titles and understand **why** certain films earn this distinction, across genres, eras, directors and countries. The findings will guide curation, "hidden gem" surfacing and editorial storytelling around the list.

2. Objective

As a data analyst on the Content Insights team, the aim is to explain **Average IMDb Rating** within the Top 700 by:

- Breaking the metric down by Genre, Decade, Director, Actor and Country.
- Finding the highest and lowest rated segments in each dimension.
- Measuring how Votes and Runtime relate to IMDb Rating.
- Testing whether director and cast track record affects rating.
- Turning the evidence into clear recommendations for the team.

3. Dataset Overview

The dataset has **700 films and 10 columns**, covering release years 1921 to 2026.

Column	Meaning	Type
Rank, Title	Position on the list and film name	Number, Text
Year	Release year (used to derive Decade)	Number
Genre(s)	One to three genres, separated by "|"	Text
Director	Director(s), separated by "|"	Text
Main Actor(s)	Leading actors, separated by "|"	Text
Country	Country codes listed for the film	Text
IMDb Rating	Bayesian-weighted score out of 10	Decimal
Votes	Number of IMDb user votes	Number
Runtime (mins)	Film length in minutes	Number

4. Key Metric and Supporting KPIs

Key metric: Average IMDb Rating is the mean of IMDb Rating for a segment (genre, decade, director or country). It shows whether a segment consistently produces enduring films or only occasionally reaches the list.

KPI	Calculation	What it shows
Average Votes per Film	Mean of Votes, by genre and decade	Whether a high rating has broad audience backing
Director Frequency	Top 700 films per director	Which directors repeatedly make list-worthy films
Country Representation Rate	Films per country / 700	Which regions are over- or under-represented

Breakdown tree: Average IMDb Rating is examined by Genre, Decade, Director, Actor, Country, and by Votes and Runtime.

5. Hypotheses

	Hypothesis
H1	Films with more votes have higher ratings
H2	Longer films have higher ratings
H3	Drama films rate higher than non-Drama films
H4	Films released before 1980 rate higher than films from 2000 onwards
H5	Directors with 4 or more Top 700 films rate higher than directors with one
H6	Films with a famous repeat actor (5+ Top 700 films) rate higher
H7	Average rating differs between countries

6. Data Inspection, Cleaning and Preparation

6.1 Structure, missing values and duplicates

- All numeric columns are stored correctly. Ratings run from **7.5 to 9.3** (mean 8.00), votes from 52 thousand to 3.2 million, and runtime from 68 to 321 minutes.
- **Missing values:** only Main Actor(s) has gaps (10 films, mostly documentaries with no cast). These were filled with "Not listed" rather than deleted.
- **Duplicates:** no repeated rows. "The Girl with the Dragon Tattoo" appears twice but is two different films (2009 and 2011), so both were kept.
- **Repeated names:** 74 films listed the same actor twice in one cell. The repeats were removed so actors are counted once.

6.2 Preprocessing steps

Action	Reason
Renamed columns to short names	Easier to code and read
Added a Decade column (1994 becomes 1990)	Group films by era
Split Genre, Director and Actor into lists, then one row per value	One film can have several of each
Took the first director as Main Director	One director per film for grouping
Min-max normalised Votes and Runtime to a 0-1 scale	Compare variables with different units

6.3 Outliers

The IQR rule flagged 15 rating outliers, 29 vote outliers and 14 runtime outliers. **All were kept.** They are genuine, not errors: the all-time classics (The Shawshank Redemption, The Godfather), the most-voted films, and real epics such as Gangs of Wasseypur (321 minutes). Removing them would remove the films the analysis is meant to explain, so averages are read with care.

Python code used:

```
Q1, Q3 = df[col].quantile(0.25), df[col].quantile(0.75)
IQR = Q3 - Q1
outliers = df[(df[col] < Q1 - 1.5*IQR) | (df[col] > Q3 + 1.5*IQR)]
```

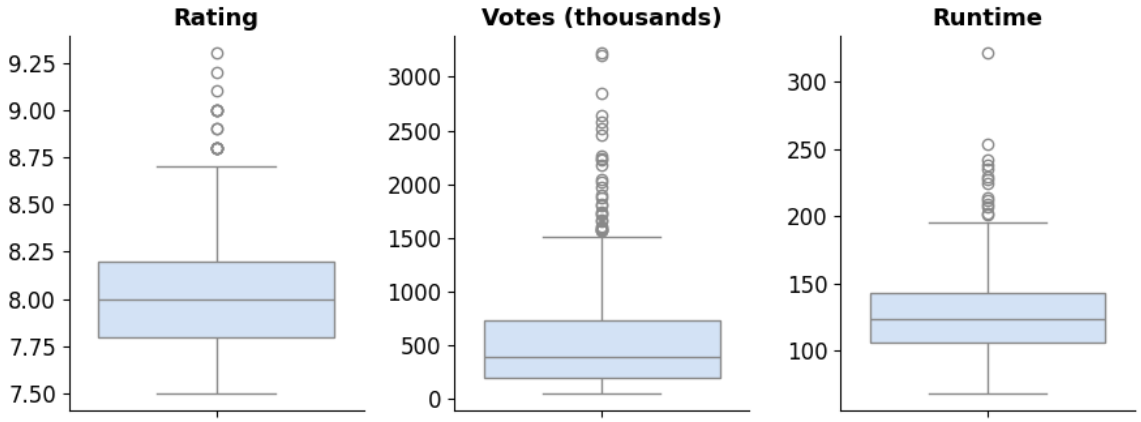


Figure 1. Box plots of Rating, Votes and Runtime

6.4 Normalisation

Votes (up to 3.2 million) and Runtime (68 to 321 minutes) are on very different scales. Both were rescaled to 0-1 using min-max normalisation, (value - minimum) / (maximum - minimum), so the most-voted film scores 1.0 and the least-voted 0.0. A Votes_K column (votes in thousands) was also added for readable charts.

```
df['Votes_Normalised'] = (df['Votes'] - df['Votes'].min()) / (df['Votes'].max() - df['Votes'].min())
```

7. Exploratory Data Analysis

7.1 Overall picture

KPI	Value
Films in list	700
Average IMDb Rating	8.00
Average Votes per Film	533,097
Average Runtime	127.4 minutes
Distinct directors	448

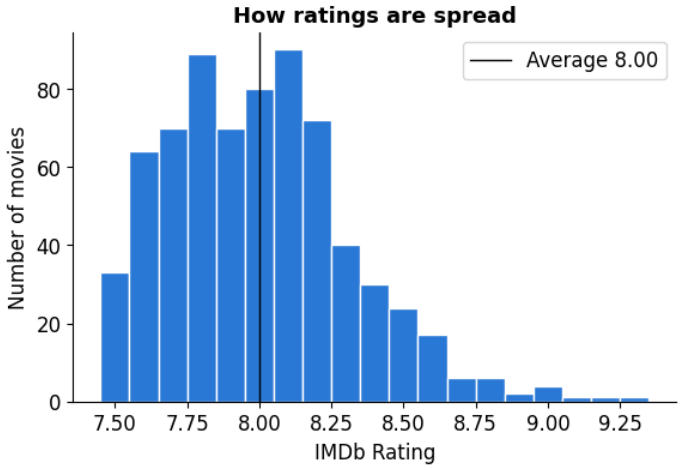


Figure 2. Distribution of ratings

Observations: the list has 700 films from 448 directors across 105 years. Average votes (533,097) are far above the median (about 399,000), so a small number of mega-popular films pull the average up. Most films sit between 7.6 and 8.2, and only a handful reach 8.8 or above.

Because every film is already strong, differences between groups are small (often 0.1 points), so statistical tests are used later to separate real patterns from noise.

7.2 Genre analysis

Approach: each genre was counted separately (a film with three genres appears three times), and only genres with at least 10 films were ranked.

```
genre_df = df.explode('Genre_List')
genre_table = genre_df.groupby('Genre').agg(Movies=('Title','count'),
          Avg_Rating=('Rating','mean'), Avg_Votes=('Votes','mean'))
genre_table = genre_table[genre_table['Movies'] >= 10]
```

Genre	Films	Avg Rating	Avg Votes
War	32	8.18	474K
Western	13	8.12	490K
Biography	78	8.04	427K
Drama	473	8.03	481K
Animation	63	8.02	501K
Crime	156	8.01	550K
Music	16	8.01	392K
Sport	16	7.99	344K
Sci-Fi	62	7.99	879K
Mystery	80	7.98	514K
Adventure	176	7.98	659K
History	27	7.97	412K
Romance	72	7.96	416K
Fantasy	57	7.96	704K
Family	36	7.95	468K
Comedy	149	7.93	479K
Action	160	7.92	595K
Thriller	106	7.92	567K
Horror	28	7.89	579K

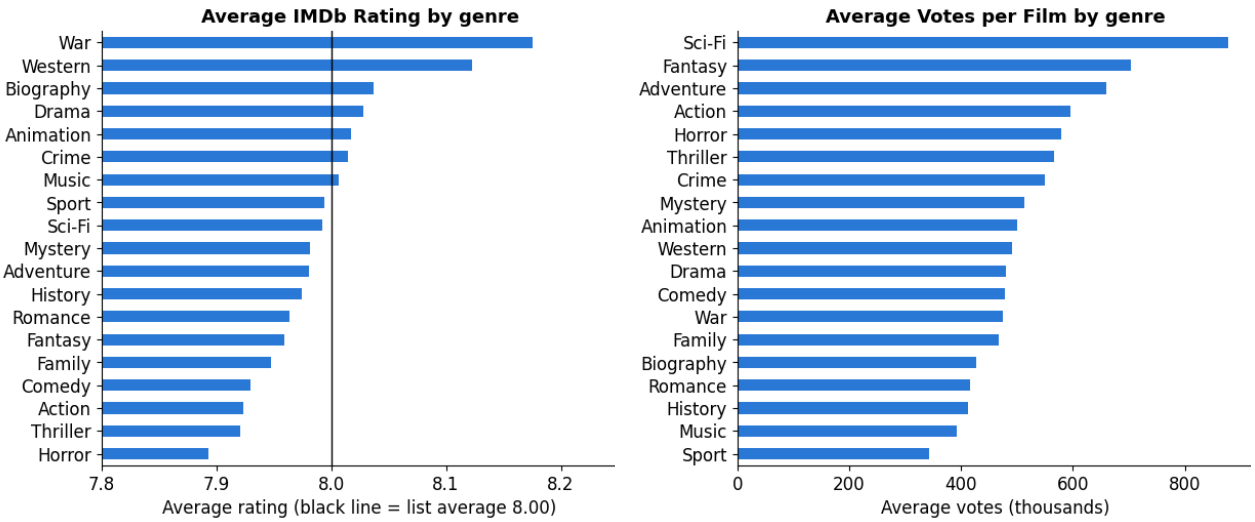


Figure 3. Average rating (left) and average votes per film (right) by genre

- **Highest rated:** War (8.18), Western (8.12), Biography (8.04), Drama (8.03). **Lowest:** Horror (7.89), Thriller and Action (7.92), Comedy (7.93).
- Drama appears in 473 of 700 films (68%), by far the most common genre.
- **Most votes per film:** Sci-Fi (879K), Fantasy (704K), Adventure (659K). Spectacle genres draw huge audiences, while serious genres earn higher ratings from smaller audiences.
- The gap between best and worst genre is only 0.29 points (8.18 vs 7.89), so genre alone is a modest predictor. War and Western have few films (32 and 13), so their averages are less stable than Drama (473).

Recommendation: feature War, Biography and Drama as "timeless picks", and use Sci-Fi and Fantasy as high-reach entry points for new viewers. Treat Western and War as small but consistently strong niches worth a dedicated editorial list.

7.3 Decade analysis

```
df['Decade'] = (df['Year'] // 10) * 10
decade_table = df.groupby('Decade').agg(Movies=('Title','count'),
        Avg_Rating=('Rating','mean'), Avg_Votes=('Votes','mean'))
```

Decade	Films	Avg Rating	Avg Votes
1920s	5	8.08	141K
1930s	8	8.18	251K
1940s	14	8.17	249K
1950s	29	8.17	239K
1960s	31	8.14	273K
1970s	34	8.09	524K
1980s	61	8.01	479K
1990s	95	8.06	702K
2000s	177	7.92	609K
2010s	181	7.93	605K
2020s	65	8.06	318K

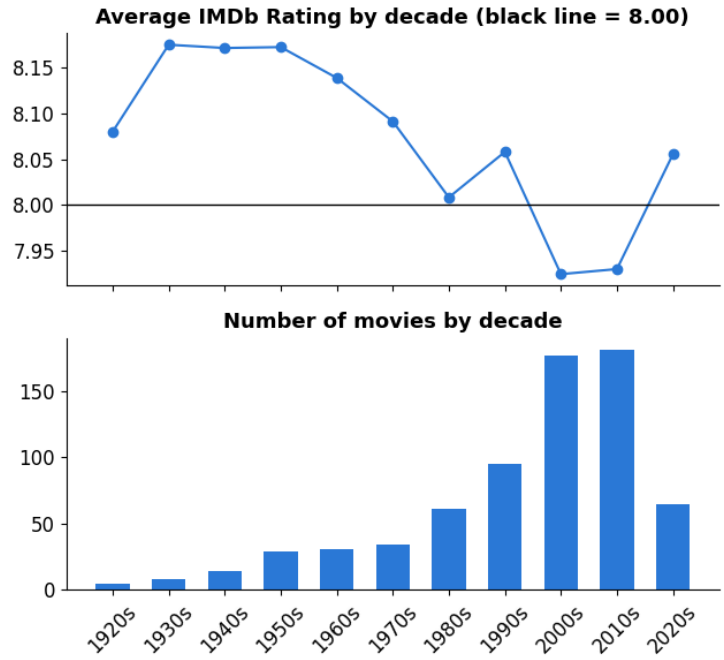


Figure 4. Average rating and number of films by decade

- **Old films rate highest:** the 1930s to 1960s average 8.14 to 8.18. Only the very best old films are still watched and voted on, so those in the list are all excellent.
- The 2000s and 2010s hold 358 films (51% of the list) but have the **lowest averages (7.92 to 7.93)**, because many "very good" modern films qualify.
- The 2020s average 8.06, but these films are new and their ratings may settle as more people vote.
- Votes per film peak in the 1990s (702K) and fall for both older films (about 240K in the 1950s) and the newest ones (318K in the 2020s), so audience size follows how long a film has been available to be discovered.

Recommendation: label 2020s films as "rising" until their ratings settle, and pair classic titles with newer, high-vote films to bridge audiences across eras.

Average Votes by Genre and Decade (supporting KPI): a pivot table of average votes (genres in rows, decades in columns) was built for the 8 largest genres from the 1950s onwards. the heatmap below shows that the 1990s to 2010s attract the most votes in almost every genre, since these are the years today's IMDb users grew up in. Older films have high ratings but far fewer votes.

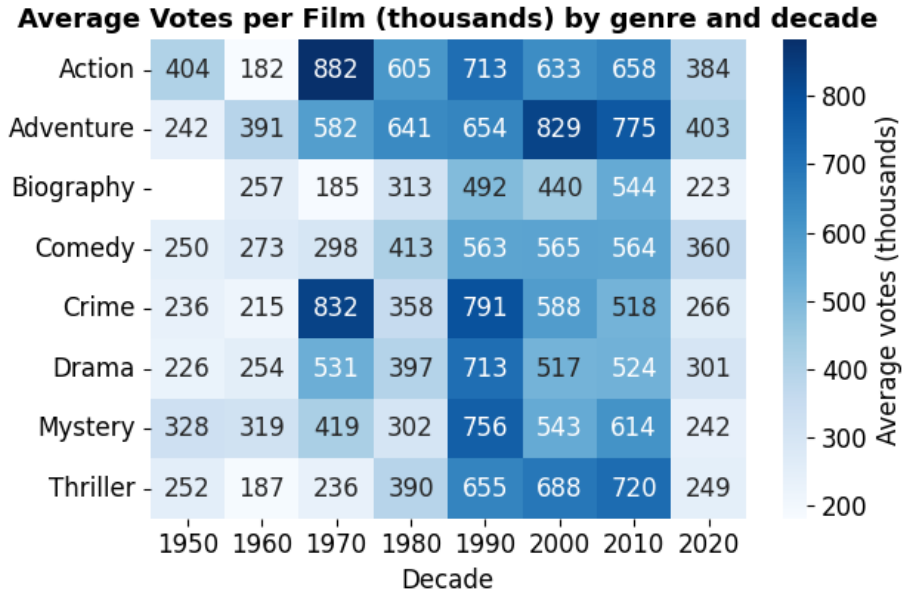


Figure 5. Average Votes per Film (thousands) by genre and decade

7.4 Director analysis

Director Frequency: 448 directors appear in the list, and **303 (68%) have only one film**. The most frequent are Steven Spielberg (12), Martin Scorsese (11), Christopher Nolan, Stanley Kubrick and Quentin Tarantino (10 each), Alfred Hitchcock (9) and David Fincher (8).

```
director_df = df.explode('Director_List')
director_counts = director_df['Director_Name'].value_counts()
df['Director_Films'] = df['Main_Director'].map(director_counts)
df['Director_Group'] = pd.cut(df['Director_Films'], [0,1,3,100], labels=['1 film','2-3 films','4+ films'])
```

Director	Top 700 films
Steven Spielberg	12
Martin Scorsese	11
Christopher Nolan	10
Stanley Kubrick	10
Quentin Tarantino	10
Alfred Hitchcock	9
David Fincher	8
Hayao Miyazaki	7
Clint Eastwood	7
Denis Villeneuve	7

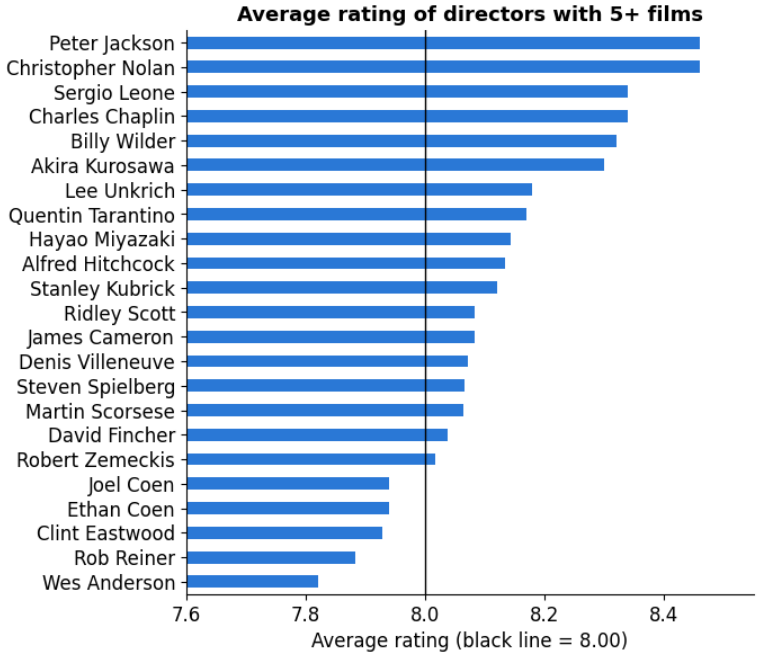


Figure 6. Average rating of directors with 5 or more films

To test track record, each film was given the number of Top 700 films by its main director and grouped:

Director group	Films	Avg Rating	Avg Votes	In Top 100
1 film	254	7.98	387K	7.5%
2 to 3 films	222	7.96	486K	10.8%
4+ films	224	8.06	745K	25.5%

- Films by directors with 4+ films average **8.06** versus 7.98 for one-time directors. The larger gap is at the very top: **25% of their films are in the Top 100, over three times the 7.5% of one-time directors.**
- Best repeat directors by average rating: Peter Jackson and Christopher Nolan (8.46), then Charles Chaplin, Sergio Leone, Billy Wilder and Akira Kurosawa (8.30 to 8.34).
- The Director Frequency KPI is highly concentrated: 303 of 448 directors (68%) appear once, while 11 directors account for 7 or more films each.

Recommendation: build director collections (Nolan, Kurosawa, Leone, Wilder, Miyazaki) because proven directors reliably deliver list-worthy work, and use "more from this director" prompts to move viewers from one acclaimed film to the next.

7.5 Actor analysis

Robert De Niro (16 films), Leonardo DiCaprio and Tom Hanks (14 each) and Al Pacino (12) appear most often. Films with a famous repeat actor (5+ films) were compared with the rest:

Actor	Top 700 films
Robert De Niro	16
Leonardo DiCaprio	14
Tom Hanks	14
Al Pacino	12
Brad Pitt	11
Tom Cruise	11
Matt Damon	11
Emma Watson	10
Christian Bale	10
Harrison Ford	9

```
actor_counts = actor_df['Actor'].value_counts()
famous = set(actor_counts[actor_counts >= 5].index)
df['Has_Famous_Actor'] = df['Actor_List'].apply(lambda a: any(x in famous for x in a))
```

Famous repeat actor	Films	Avg Rating	Avg Votes
No	460	8.00	420K
Yes	240	8.00	750K

Famous actors bring **bigger audiences (750K vs 420K votes) but no higher rating (8.00 vs 8.00)**. Cast fame drives reach, not quality.

7.6 Country analysis

Listed main country	Films	Share	Avg Rating
Denmark	378	54.0%	8.03
Latvia	189	27.0%	7.92
Philippines	69	9.9%	7.98
Austria	26	3.7%	8.03
Portugal	10	1.4%	8.00
South Africa	10	1.4%	8.22

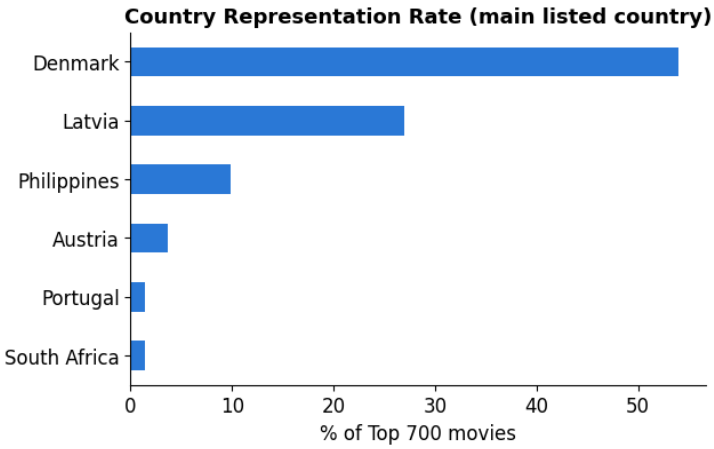


Figure 7. Country Representation Rate (main listed country)

By the listed codes, Denmark accounts for 54% of films, Latvia 27% and the Philippines 10%. However, checking known films shows these codes are wrong: The Shawshank Redemption (American) is listed as Denmark and Austria, Parasite (Korean) as Denmark and Latvia, and 3 Idiots (Indian) as Latvia and the Philippines. **The Country column does not show where films were made.** Country results are therefore not reliable for decisions until the column is corrected. This is an important data-quality finding in its own right.

Recommendation: rebuild the Country field from each film's production countries and re-run this analysis. The Country Representation Rate KPI can only be interpreted once codes map to real origins, for example to see whether Indian, Korean or Japanese cinema is under-represented relative to its global output.

7.7 Votes and runtime versus rating

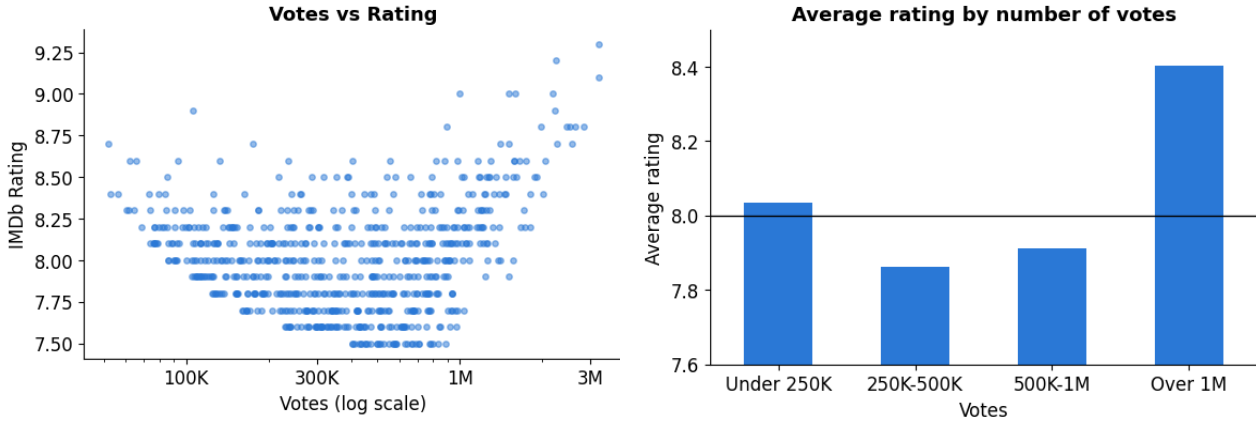


Figure 8. Votes vs rating (log scale, left) and average rating by vote group (right)

Vote group	Films	Avg Rating
Under 250K	232	8.04
250K to 500K	187	7.86
500K to 1M	195	7.91
Over 1M	86	8.40

- Correlation between Votes and Rating is **+0.40** (moderate). The vote groups reveal a **U shape**: films above 1M votes average 8.40 (the famous classics, 79% of them in the Top 100), films under 250K votes average 8.04, and the middle groups are lowest.
- **Why the U shape?** Under the weighted formula, a film with few votes is pulled towards the average, so it only enters the Top 700 if its rating is exceptional. Low-vote films in the list are therefore very strong.
- **Hidden gems:** 55 films have a rating of 8.2 or more with fewer than 200,000 votes, for example Punjab '95, 12th Fail, Harakiri, Soorarai Pottru, Sita Ramam and Ikiru.

Hidden gem	Year	Rating	Votes
Punjab '95	2026	8.9	105,726
12th Fail	2023	8.7	175,277
Kill Bill: The Whole Bloody Affair	2004	8.7	51,964
Harakiri	1962	8.6	93,058
20 Days in Mariupol	2023	8.6	65,359
Rocketry: The Nambi Effect	2022	8.6	62,428
Soorarai Pottru	2020	8.6	132,475
Sita Ramam	2022	8.5	84,938

```
df['Hidden_Gem'] = (df['Rating'] >= 8.2) & (df['Votes'] < 200_000)
```

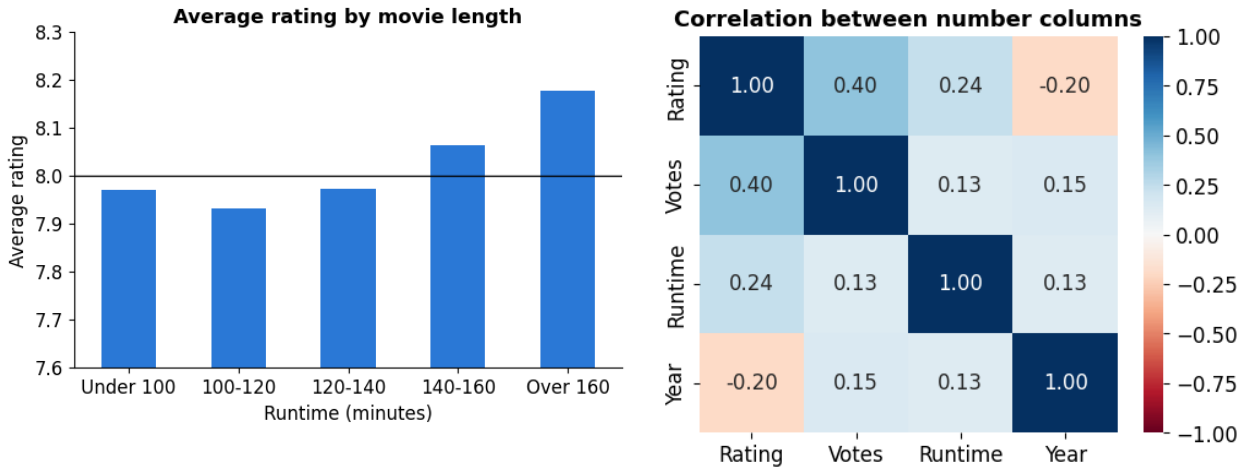


Figure 9. Average rating by runtime group (left) and correlation heatmap (right)

Runtime (mins)	Films	Avg Rating
Under 100	102	7.97
100 to 120	202	7.93
120 to 140	200	7.97
140 to 160	105	8.06
Over 160	91	8.18

- Runtime and Rating correlate at **+0.24** (weak positive). Films over 160 minutes average 8.18, against 7.93 for 100 to 120 minutes. Long epics such as Lawrence of Arabia, Schindler's List and The Godfather Part II are among the best loved.
- The correlation heatmap confirms that all links are modest. No single numeric variable explains rating on its own, which is why genre, era and director were examined alongside votes and runtime.
- Across all numeric columns: Rating and Votes +0.40, Rating and Runtime +0.24, Rating and Year -0.20 (newer films rate slightly lower).

8. Hypothesis Testing

Group averages were compared with Welch's t-test (two groups) and ANOVA (more than two). A p-value below 0.05 means the difference is unlikely to be chance. H1 and H2 were assessed with correlation.

```
from scipy import stats
stats.ttest_ind(group_a, group_b, equal_var=False)   # H3 to H6
stats.f_oneway(*country_groups)                      # H7 (countries with 10+ films)
```

	Hypothesis	Evidence	Result
H1	More votes, higher rating	Correlation +0.40; U-shaped vote groups	Partly true
H2	Longer films rate higher	Correlation +0.24; 8.18 for 160+ min	True
H3	Drama rates higher	8.03 vs 7.94, $p = 0.0006$	True
H4	Older films rate higher	8.14 vs 7.95, $p < 0.0001$	True
H5	Directors with 4+ films rate higher	8.06 vs 7.98, $p = 0.0065$	True
H6	Famous actors raise rating	8.00 vs 8.00, $p = 0.7924$	False
H7	Rating differs by country	ANOVA $p = 0.0008$	True, but data unreliable

- **H1 (partly true):** votes and rating are positively related, but not in a straight line. Both the most-voted and the least-voted films rate high, while mid-range films rate lowest.
- **H2 (true):** runtime has a weak positive link with rating, driven by long epics.
- **H3 (true):** Drama films average 8.03 against 7.94 for the rest.
- **H4 (true):** pre-1980 films average 8.14 against 7.95 for 2000 onwards.
- **H5 (true):** directors with a strong track record show a significant, though small, rating advantage and a much larger Top 100 advantage.
- **H6 (false):** with an identical 8.00 average and $p = 0.79$, a famous actor makes no difference to rating.
- **H7 (true, unreliable):** the differences are significant, but the Country column is flawed, so this should not guide decisions.

Reading the results: the effects are small in absolute terms (ratings span only 7.5 to 9.3) and these are associations, not proof of cause. H4 is also affected by survivorship, since only the best older films remain widely voted on.

9. Key Insights

1. **Votes matter, in a U shape.** The most-voted films (1M+) are true classics (8.40 average), while low-vote films earn their place only with very high ratings.
2. **Serious genres rate highest.** War, Western, Biography and Drama lead; Horror, Thriller, Action and Comedy trail. Sci-Fi, Fantasy and Adventure attract the most votes.
3. **Old classics rate highest.** The 1930s to 1960s average about 8.15, while the crowded 2000s and 2010s average about 7.93.
4. **Longer films rate higher.** Films over 160 minutes average 8.18.
5. **Great directors repeat their success.** Directors with 4+ films are over three times more likely to reach the Top 100.
6. **Famous actors bring votes, not ratings.**
7. **The Country column has errors** and needs fixing before regional analysis.

10. Recommendations

	Recommendation	Based on
1	Create a "Hidden Gems" section with the 55 high-rated, low-vote films	Insight 1
2	Build director collections (Nolan, Kurosawa, Leone, Wilder, Miyazaki)	Insight 5
3	Promote classics and War, Biography and Drama films as "timeless picks"	Insights 2, 3
4	Label 2020s films as "rising" until their ratings settle	Decade analysis
5	Use famous actors to attract viewers, not as a quality signal	Insight 6
6	Fix the Country column before any country-based reporting	Insight 7

11. Tableau Dashboard

The findings were also built into an interactive Tableau workbook with a 5-point storyboard titled "Decoding What Makes a Movie Timeless". It is organised in two layers: a main dashboard where users can explore the data with filters, and a storyboard that walks a stakeholder through the key messages.

11.1 Main dashboard: "IMDb Top 700: What Makes a Movie Timeless?"

- **KPI cards:** Total Movies (700), Average IMDb Rating (8.00), Average Votes (533K) and Average Runtime (127 minutes). The cards update whenever a filter is applied.
- **Eight linked charts:** Average IMDb Rating by Genre, Votes by Genre, Rating by Decade (line, with film counts), Top Directors (director track record), Rating by Runtime group, Rating by Vote group, Votes vs Rating (one point per film), and Country Representation.
- **Filters:** Genre, Decade, Vote Group and Director Group, so a viewer can, for example, look only at Drama films from the 1990s or only at directors with 4+ films.
- **Interactivity:** clicking a decade or a film in the Votes vs Rating chart filters the rest of the dashboard, and films flagged as hidden gems are highlighted by colour.
- **Data-quality note:** the Country chart carries a caption stating that the country codes are as given in the data and do not show where films were made, matching the finding in Section 7.6.

11.2 Storyboard

Story point	Message shown	Report section
1	700 movies, average rating 8.00: what do the best movies have in common?	4, 7.1
2	War, Western, Biography and Drama rate highest; Horror and Action rate lowest	7.2
3	Old classics rate highest; the busy 2000s and 2010s rate lowest	7.3
4	Great directors repeat their success: 4+ films means 3x more likely to be in the Top 100	7.4
5	55 hidden gems: high rating, few votes. Recommend them!	7.7, 10

Each story point corresponds to one key insight in this report, so the dashboard and the report tell the same story. The workbook uses the cleaned data and derived fields prepared in the notebook (Decade, Main Director, Director Group, Vote Group, Runtime Group and Hidden Gem).

Dashboard link:
https://public.tableau.com/views/NL_IMDbTop700_17912856364920/DecodingWhatMakesaMovieTimeless

12. Limitations

- Every film is already in the Top 700, so the analysis explains differences **within** a high-quality group, not what separates good films from poor ones.
- Averages for small groups (Western, 13 films; 1920s, 5 films) are less reliable than those for large groups.
- Multi-genre films are counted once per genre, so genre counts add up to more than 700.
- Only the first listed director is used for grouping, which can understate co-directed films.

13. Conclusion

A film's lasting reputation in the Top 700 is linked to serious, character-driven genres, older release dates, longer runtimes and a proven director, and is backed by either very broad audience consensus (1M+ votes) or an exceptionally high score from a smaller audience. Cast fame raises reach but not ratings. The Country column should be corrected before regional conclusions are drawn.